

Cyber Security Foundations Guide

Vulnerability Scanning aur Security Concepts ka Basic Overview

Aapke dwara diye gaye cyber security ke mahatvapurna points ko ek vyavasthit aur professional format mein niche prastut kiya gaya hai. Isko padhakar aap security assessment ke buniyadi tariko ko samajh sakte hain.

1. Vulnerability Scanning Kya Hai?

System ki security kamiyon (vulnerabilities) ko automatic tarike se dhundhne ki process ko **Vulnerability Scanning** kehte hain. Yeh tools ke madhyam se poore network ya website ka automatic audit karta hai taaki kamzor kadiyon ka pata lagaya ja sake.

2. Difference: Vulnerability Scanning vs Penetration Testing

Dono hi security check karne ke tarike hain, lekin inke kaam karne ke tarike mein bada antar hota hai:

Feature / Antar	Vulnerability Scanning	Penetration Testing
Prakriya (Process)	Yeh poori tarah se Automated (tool-based) hoti hai.	Yeh adhitam Manual aur complex hoti hai.
Karya (Action)	Yeh sirf kamiyon (vulnerabilities) ko dhoondh kar list banati hai.	Isme kamiyon ko dhoondhne ke sath-sath Actual Attack Simulation karke access lene ki koshish ki jaati hai.
Samay aur Skill	Isme kam samay lagta hai aur basic tools ki knowledge kaafi hai.	Isme gehra technical gyan (expert skill) aur zyada samay lagta hai.

3. Common Vulnerabilities (Aam Kamiyan)

Aksar servers aur web applications mein niche di gayi aam security kamiyan milti hain:

- **Outdated Software:** Purane software ya operating system versions ka upyog karna jinme pehle se known bugs hon.
- **Weak Passwords:** Default ya kamzor administrative passwords rakhna jinhe aasani se guess ya brute-force kiya ja sake.

- **Misconfigured Services:** Server ki galat settings, jaise database port (3306) ya FTP port ka public internet par open reh jana.

4. Scanners Kaise Detect Karte Hain?

Automated scanners target system par chal rahi active services aur unke versions ko identify karte hain. Iske baad, scanners un versions ko ek standard global database jaise **CVE (Common Vulnerabilities and Exposures)** se match karte hain. Agar database mein us version ke liye koi kami milti hai, to scanner use report mein dikha deta hai.

5. CVSS (Common Vulnerability Scoring System) Kya Hai?

CVSS Score Definition

Yeh ek standard scoring system hai jo kisi bhi dhoondhi gayi vulnerability ki severity (risk level) ko 0 se 10 tak ke aadhar par ek score deta hai. Isse pata chalta hai ki nuksandeh kami kitni gambhir hai:

Low: 0.1 - 3.9

Medium: 4.0 - 6.9

High/Critical: 7.0 - 10.0

6. Frequency of Scans (Scans Kab Karne Chahiye?)

Vulnerability assessment ek baar karne wali cheez nahi hai. Isko niyमित रूप से किया जाना चाहिए:

- **Regular Schedule:** Har hafte (Weekly) ya har mahine (Monthly) regular scans run karne chahiye.
- **Event-Based:** Jab bhi system mein koi naya bada update kiya jaye, naya infrastructure add ho, ya koi naya software patch lagaya jaye, tab scan karna zaroori hai.

7. False Positive Kya Hota Hai?

Jab koi scanner kisi poori tarah se secure service ya configuration ko galti se khatra (vulnerability) dikha de, to use **False Positive** kehte hain. Yeh tools ki ek limitation hoti hai jise baad mein security analysts manually verify karke filter karte hain.

8. Prioritization (Kamion ko Theek Karne ka Krama)

Saari vulnerabilities ko ek sath theek karna namumkin hota hai. Isliye, team **CVSS Score** aur us kami ke business/system par padne wale **actual impact (prabhav)** ke aadhar par priority set karti hai. Sabse pehle critical aur high scores wali kamion ko patch kiya jata hai.